physics modelling agent

progress report

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outline

aim of the project

what the “physics modelling agent” should be

A configuration of “intelligent systems” which assists a human researcher in building a model to answer a research question.

how the model should work

human input: free form text modelling request

“physics modelling agent” takes the input:
- builds a model
- crafts numerical implementation of the model

reports findings:
- human readable **model description**
- executable **numerical implementation**

definitions of LLM and agentic AI

what is an LLM

- "Transformer language models" [19] with at least billions of parameters (no clear cut-off) [19]
- mostly use tokenization: dividing text into pieces and applying transformation function

what is an AI

- general: system that acts to achieve the best version of the expected result [15]

what is an agentic AI

- an AI with enhanced capabilities including: strategic planning, (external) tools usage, collaboration with other agents [3]

architectural foundations

constraints

- only use open source software
- try to use only free software (no payed subscriptions)

design choices

- use agentic AI to harness strategic planning capabilities and tool usage

→ Scalable AI Accelerator (**SAIA**) from the GWDG[10]
  - access via remote servers with valid API key [9]
- use an existing framework for AI orchestration
  - save time on the software engineering part of the project
  - focus on definition of AI agents and configuration of their behaviour

Figure: framework chosen: crewai [6]

crewai agents

Agents[4]

accesses an agentic AI model to perform a task

- allows to access any AI-models we have access to (SAIA)
- behaviour can be defined via parameters (role, goal, backstory, creativity)
- tools can be added to enhance capabilities (e.g. web search, code execution)

Tasks[7]

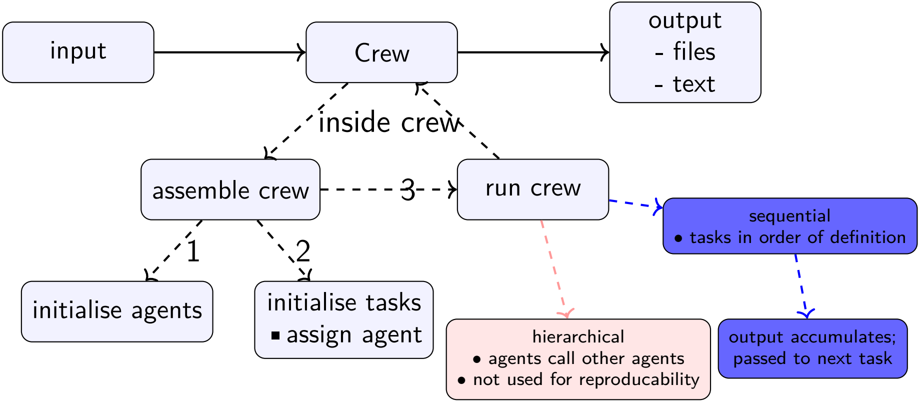
describe a concrete action to execute

- defined by task description and expected output
- is assigned an agent (execution merges agent and task description)

crewai: orchestration

the Crew[5]

contains all agents with their tasks and handles the interactions between agents



orchestrating agentic AI's

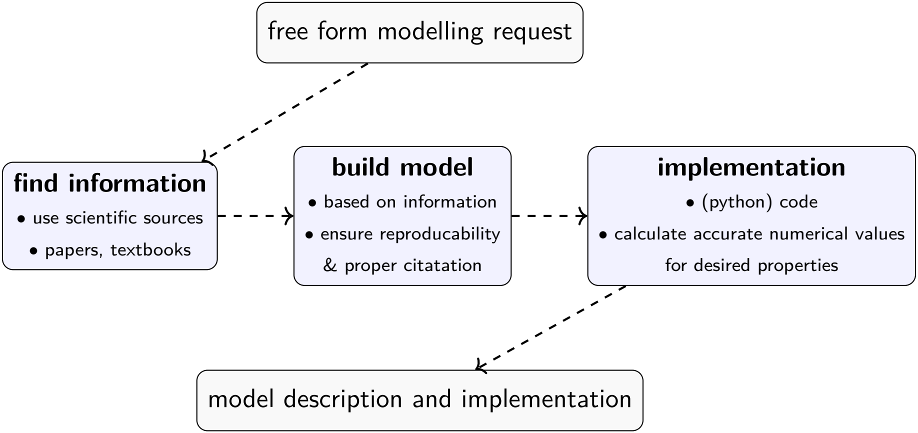


Figure: general vision for the steps the crew should perform

information finding

problems encountered

1. crewai agent on it's own can not perform a live web-search

✓ use arXiv API Access[2]; scientific sources available

2. LLM models with strong reasoning capabilities can not read PDFs

✓ split this step into two tasks

   A find and download papers via arXiv API

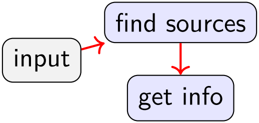
   B have a capable agent extract relevant information from the downloaded papers

3. crewai tool gets denied by arXiv servers due to too many requests

✓ implement custom tools performing exponential back-off

✓ separate the task performing the arXiv request from the crew

$\Longrightarrow$ no new requests when testing other tasks



modelling

the unit problem

1. task outputs formula from the sources; theoretical formulas often use different definitions of parameters

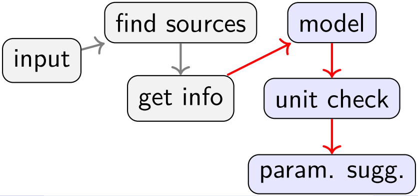
✓ custom tool using SymPy [17]:

   A perform dimensional analysis to find possible unit mismatches
   B correct the formulas base on dimensional analysis

starting parameters

- for realistic results realistic start parameters are necessary

$\Longrightarrow$ agent for suggesting reasonable starting parameters



numerical implementation

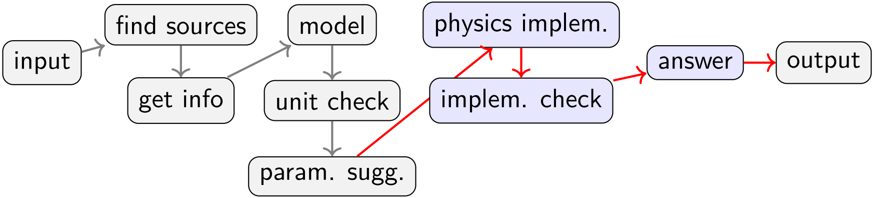
minor challenges

1. first simulation agent specialized on logic often puts out code with bugs

✓ second agent used LLM specialized in programming and corrects these mistakes

answer: provide a short result at runtime

- needed for automatic evaluation of the system (benchmarking)



example: input/modelling request

input

aim : “Calculate the Edelstein effect for a Rashba fermion (at the Gamma point of the Brillouin zone).
Compute the magnitization magnitude and direction of different directions and magnitudes of the applied electric field.
Consider how the result depends on relevant parameters of the model (e.g. chirality, fermi velocity, spin-orbit coupling strength) and make explicit graphics.”

topic : Edelstein-Effect

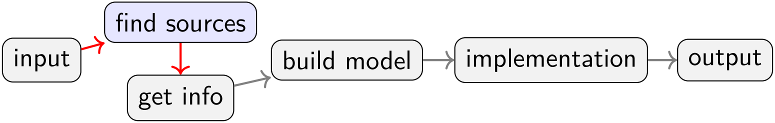
papers found

agent configuration:

- no creativity (temperature = 0)
- LLM model: qwen3.5-122b-a10b[16] (general purpose & usage)
- tools: arXiv_Papers_With_Backoff

papers found (view full report)

"Edelstein Effect in Isotropic and Anisotropic Rashba Models"[12]
"Spin and orbital Edelstein effect in spin-orbit coupled noncentrosymmetric superconductors"[1]
"Spin and orbital Edelstein effect in a bilayer system with Rashba interaction"[14]



information extraction

agent configuration:

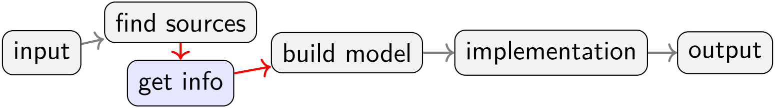
- no creativity (temperature $= 0$)
- LLM model: qwen3.5-122b-a10b[16] (general & tool usage)
- tools: pDF_Reader

extracted information (view full report)

$$\hat{H} = \frac{p^2}{2m} + \alpha \hat{z} \cdot (\vec{p} \times \vec{\sigma})$$ Rashba-Hamiltonian

$$M_y = \frac{\mu_b |e| \tau}{2\pi} m\alpha \left[\hat{z} \times \mathbf{E}\right]_y$$ magnetization in HDR

$$M_y = \frac{\mu_b |e| \tau}{2\pi} \sqrt{m^2\alpha^2 + 2mE_F} \left[\hat{z} \times \mathbf{E}\right]_y$$ magnetization in LDR

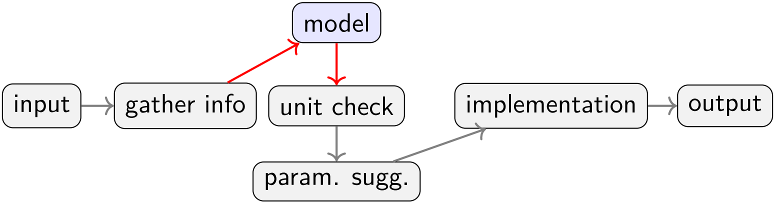


agent configuration:

- little creativity (temperature $= 0.2$)
- LLM model: deepseek-r1-distill-llama-70b[13]; (optimized for logic & reasoning)
- tools: this LLM is unable to use tools

extracted information (view full report)

restates results from information extraction



unit check

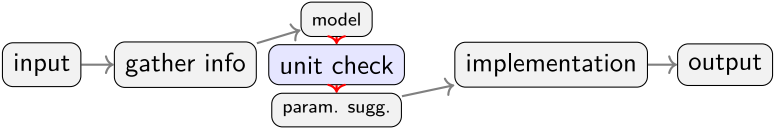
- little creativity (temperature $= 0.2$)
- LLM model: qwen3.5-122b-a10b[16]; (allow tool usage)
- tools: dimensional_Analysis using SymPy [17]

unit corrections (view full report)

$$\hat{H} = \frac{p^2}{2m} + \frac{\alpha}{\hbar}\hat{z}\cdot(\vec{p}\times\vec{\sigma})$$ Rashba-Hamiltonian

$$M_y = \frac{\mu_b|e|\tau}{2\pi\hbar^2} m\alpha\left[\hat{z}\times \mathbf{E}\right]_y$$ magnetization in HDR

$$M_y = \frac{\mu_b|e|\tau}{2\pi\hbar^2}\sqrt{m^2\alpha^2 + 2mE_F}\left[\hat{z}\times \mathbf{E}\right]_y$$ magnetization in LDR

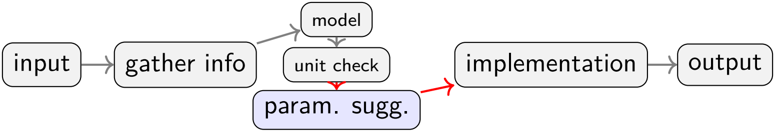


parameter suggestion

- no creativity (temperature = 0)
- LLM model: qwen3.5-122b-a10b[16]; (general purpose)
- tools: no tool used

suggested starting parameters (view full report)

$m \approx 6,10 \times 10^{-32}\text{kg}$	effective mass
$\alpha = 5,0 \times 10^{4}\text{m/s}$	Rashba Coupling Strength
$\tau = 5,0 \times 10^{-13}\text{s}$	transport time
$E_x = 1,0 \times 10^{5}\text{V/m}$	applied electric field

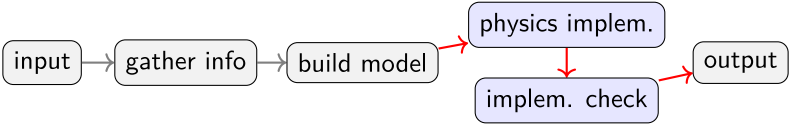


numerical implementation I

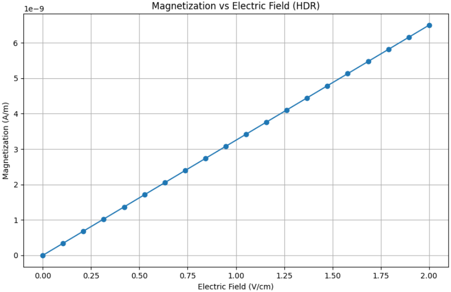
- no creativity (temperature $= 0$)
- LLM model: deepseek-r1-distill-llama-70b [13]; devstral-2-123b-instruct-2512 [11]
- tools: no tool used

results for magnetisation (view full report)

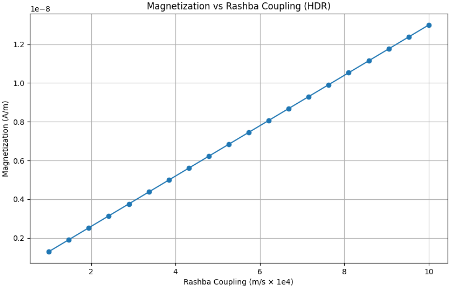
Isotropic HDR Magnetization:	$M_y = 3,25 \times 10^{-9}$ A/m
Isotropic LDR Magnetization:	$M_y = 4,65 \times 10^{-35}$ A/m
Anisotropic Magnetization Components:	$M_x = 2,04 \times 10^{-8}$ A/m
	$M_y = 2,04 \times 10^{-8}$ A/m



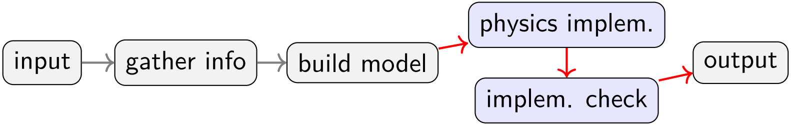
numerical implementation II



(a) magnetisation over electric field



(b) magnetisation over Rashba param. α



next steps

benchmark the "physics-modelling-agent"

a method to measure the systems performance

- use sciBench [18]: problems, solutions and execution script are open source; the problems are at high-school level
- alternatively CritPt [20][8]: evaluation only via their servers, solutions are not public; problems are more difficult

research the influence of (some of) the following aspects

1. agents temperature: which temperature maximizes performance?
2. LLM models: does performance increase with LLM complexity?
3. output variance: how consistent is the system over multiple runs?
4. execution time: how is it influenced by parameters?
5. sources: how does the number (and possibly type) of sources influence the systems results?

summary

TASK
- description
- expected output
- agent

AGENT
- role, goal, backstory
- LLM model, temperature
- assigned tools

- Input → free form modelling request
- gather information → arXiv search & information extraction
- suggest a model → based on given information
- check & correct units → dimensional analysis
- implement model → executable python code calculations

sources I

[1] Satoshi Ando et al. "Spin and orbital Edelstein effect in spin-orbit coupled noncentrosymmetric superconductors." In: *Physical Review B* 110.18 (11/2024). ISSN: 2469-9969. DOI: 10.1103/physrevb.110.184503. URL: http://dx.doi.org/10.1103/PhysRevB.110.184503.

[2] arXiv. *arXiv API Access*. 07/09/2026. URL: https://info.arxiv.org/help/api/index.html.

[3] Ajay Bandi et al. "The Rise of Agentic AI: A Review of Definitions, Frameworks, Architectures, Applications, Evaluation Metrics, and Challenges." In: *Future Internet* 17.9 (2025). ISSN: 1999-5903. DOI: 10.3390/fi17090404. URL: https://www.mdpi.com/1999-5903/17/9/404.

[4] crewai. *core.py*. 04/22/2026. URL: https://github.com/crewAIInc/crewAI/blob/main/lib/crewai/src/crewai/agent/core.py (visited on 04/22/2026).

sources II

[5] crewai. *crew.py*. 04/22/2026. URL: https://github.com/crewAIInc/crewAI/blob/main/lib/crewai/src/crewai/crew.py (visited on 04/22/2026).

[6] crewai. *crewAi*. 04/22/2026. URL: https://crewai.com/ (visited on 04/22/2026).

[7] crewai. *task.py*. 04/22/2026. URL: https://github.com/crewAIInc/crewAI/blob/main/lib/crewai/src/crewai/task.py (visited on 04/22/2026).

[8] critPt. *challanges*. 04/22/2026. URL: https://github.com/CritPt-Benchmark/CritPt/tree/main/data/public_test_challenges (visited on 04/22/2026).

[9] Gesellschaft für wissenschaftliche Datenverarbeitung mbH Göttingen. *AcademicCloud*. 04/21/2026. URL: https://id.academiccloud.de (visited on 04/22/2026).

sources III

[10] Gesellschaft für wissenschaftliche Datenverarbeitung mbH Göttingen. *SAIA*. 04/21/2026. URL: https://docs.hpc.gwdg.de/services/ai-services/saia/index.html (visited on 04/22/2026).

[11] Hugging Face. *Devstral 2 123B Instruct 2512*. 04/22/2026. URL: https://huggingface.co/mistralai/Devstral-2-123B-Instruct-2512 (visited on 04/22/2026).

[12] Irene Gaiardoni et al. "Edelstein Effect in Isotropic and Anisotropic Rashba Models." In: *Condensed Matter* 10.1 (03/2025), p. 15. ISSN: 2410-3896. DOI: 10.3390/condmat10010015. URL: http://dx.doi.org/10.3390/condmat10010015.

[13] Daya Guo et al. "DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning." In: *Nature* 645.8081 (2025), pp. 633–638. ISSN: 1476-4687. DOI: 10.1038/s41586-025-09422-z. URL: http://dx.doi.org/10.1038/s41586-025-09422-z.

sources IV

[14] Sergio Leiva-Montecinos et al. "Spin and orbital Edelstein effect in a bilayer system with Rashba interaction." In: *Physical Review Research* 5.4 (12/2023). ISSN: 2643-1564. DOI: 10.1103/physrevresearch.5.043294. URL: http://dx.doi.org/10.1103/PhysRevResearch.5.043294.

[15] Stuart Russell and Peter Norvig. *Künstliche Intelligenz Ein moderner Ansatz*. Pearson Benelux B.V. (Zweigniederlassung Deutschland), 2023. ISBN: 9783868944303. URL: https://elibrary.pearson.de/book/99.150005/9783863263263.

[16] Qwen Team. *Qwen3.5-Omni Technical Report*. 2026. arXiv: 2604.15804 [cs.CL]. URL: https://arxiv.org/abs/2604.15804.

[17] SymPy Development Team. *Welcome to SymPy's documentation!* 04/27/2025. URL: https://docs.sympy.org/latest/index.html (visited on 07/09/2026).

sources V

[18] Xiaoxuan Wang et al. "SciBench: Evaluating College-Level Scientific Problem-Solving Abilities of Large Language Models." In: *Proceedings of the Forty-First International Conference on Machine Learning*. 2024.

[19] Wayne Xin Zhao et al. *A Survey of Large Language Models*. 2026. arXiv: 2303.18223 [cs.CL]. URL: https://arxiv.org/abs/2303.18223.

[20] Minhui Zhu et al. *Probing the Critical Point (CritPt) of AI Reasoning: a Frontier Physics Research Benchmark*. 2025. arXiv: 2509.26574 [cs.AI]. URL: https://arxiv.org/abs/2509.26574.